National University of Computer and Emerging Sciences
Karachi Campus

Linear Algebra (MT1004)

Sessional-II Exam

Date: April 1st 2024

Course Instructor(s)

Mr. Muhammad Jamilusmani

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 2

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Roll No Section Student Signature

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Attempt all the questions.

CLO2 #: Understanding the core concepts of Euclidean vector spaces and matrix transformations

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Q1(a) Determine the following vectors are linearly independent or dependent in $\mathbb{R}^3$ [5]

$$v_1 = (-3, 0, 4), v_2 = (5, -1, 2\ ), v_3 = (1,1,3)$$

(b) Find the basis for the solution space of the homogenous linear system and find the dimension [5]

$$3x_1 + x_2 + x_3 + x_4 = 0$$
$$5x_1 - x_2 + x_3 - x_4 = 0$$

(c) **:** Consider the matrix $A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$ [10]

i. Convert matrix A into reduced row echelon form.
ii. Write the general solution and find the Rank and nullity of A.
iii. Find a basis for the null space of A.

(d) Consider the matrix $B = \begin{bmatrix} -14 & 12 \\ -20 & 17 \end{bmatrix}$ [10]

i. What is the characteristic equation of matrix B.
ii. Determine the eigenvalues and eigenvectors of given matrix.
iii. Find a matrix P that diagonalizes B and compute the matrix B^9